

What made Music popular?

A Retrospective of the last 20 Years

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Overview

- 1 Motivation
- 2 What we wanted to know
- 3 How we found out
 - Data Sources
 - Data Collection
 - Challenges
- 4 What we found out
 - Results
- 5 References

“I think music in itself is healing.
It's an explosive expression of humanity.
It's something we are all touched by.
No matter what culture we're from,
everyone loves music.” – *Billy Joel*

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What we wanted to know I

Generally

How did various musical/lyrical elements vary in the most popular songs of the last 20 years?

What we wanted to know II

Subquestions

- Rhythm
 - ▶ Duration
 - ▶ Tempo
- Chords
 - ▶ Chord-progressions
 - ▶ Chords and chord types
 - ▶ Intervals between chords
- Lyrics
 - ▶ Words and combinations
 - ▶ Polarity

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- **Main source:** *Billboard Year-End Hot 100 chart* [1] of 2005 up to 2024
- **Refined sources using songs from the main source:**
 - ▶ Rhythm: *Spotify API* [5], *SongBPM* [4]
 - ▶ Chords: *Ultimate Guitar* [6]
 - ▶ Lyrics: *LastFM API* [3], *Genius API* [2]

Data Collection

- Rhythm

- 1 Retrieve duration [5] and tempo [4]
- 2 Create tables

- Chords

- 1 Use chord-link finder to search for links on *Ultimate Guitar* [6]
- 2 Run chord webscraper for found song links
- 3 Create tables
- 4 Scrape genres from *LastFM API* [3] and create table

- Lyrics

- 1 Retrieve links to lyrics from *Genius API* [2]
- 2 Scrape lyrics and create tables

Challenges

- No public APIs
- Licensing restrictions
- Incorrect search results
- URL changes
- Inconsistent artist names
 - ▶ Featured artists
 - ▶ Special characters

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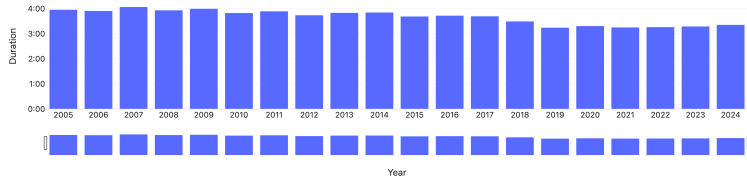


Figure: Average song duration over the years

Rhythm II

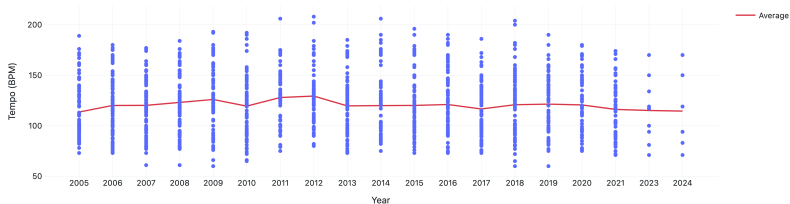


Figure: Average song tempo over the years

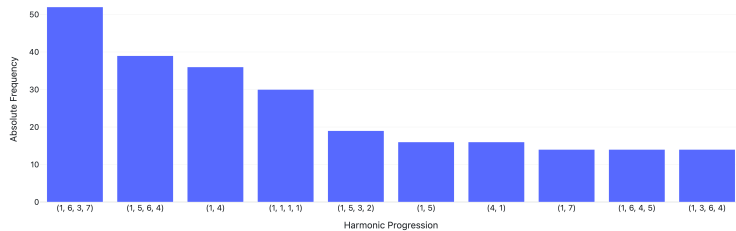


Figure: Top ten most used harmonic progressions by absolute frequency

Chord II

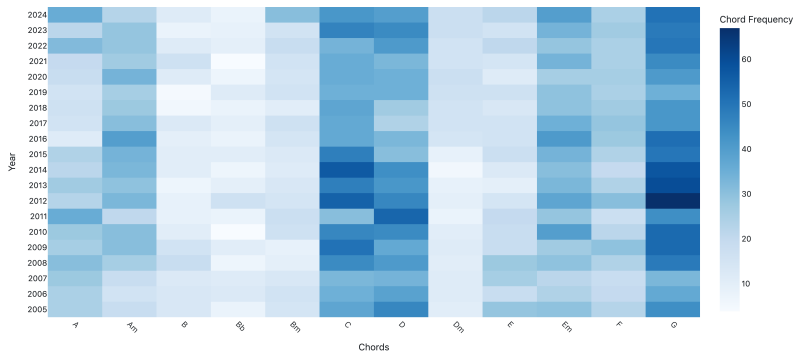


Figure: Top ten most used basic chords over the years

Chord III

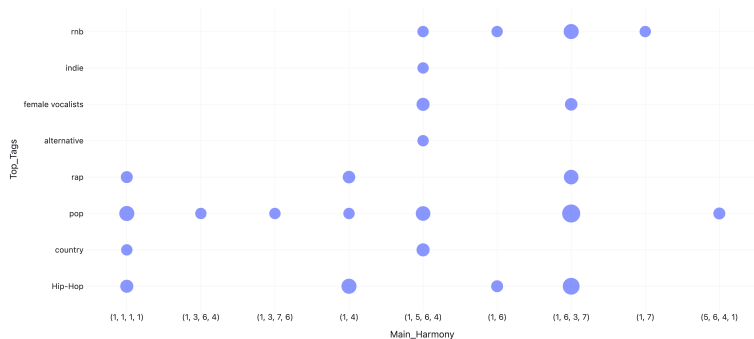


Figure: Absolut usage of chord progressions over top tags

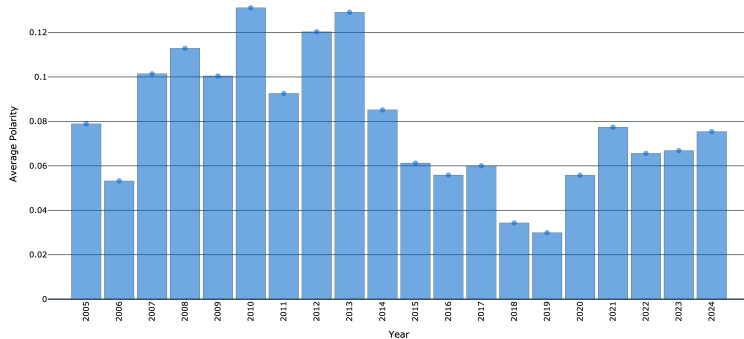


Figure: Lyric polarity over time



Figure: Wordcloud of most frequent and important words of all lyrics

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- [1] *Billboard Year-End Hot 100 chart*. URL: https://en.wikipedia.org/wiki/Category:Lists_of_Billboard_Year-End_Hot_100_singles.
- [2] *Genius API*. URL: <https://api.genius.com>.
- [3] *LastFM API*. URL: <https://www.last.fm/api>.
- [4] *SongBPM*. URL: <https://songbpm.com/>.
- [5] *Spotify API*. URL: <https://developer.spotify.com/documentation/web-api/reference/search>.
- [6] *Ultimate Guitar*. URL: <https://www.ultimate-guitar.com/>.